Synchronized speculative positioning in futures markets: a pre-registered null and exploratory sub-market effects

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Abstract

Coupled-oscillator (Kuramoto) measures are proposed as crowding indicators: when traders' position cycles align, synchronized exits should follow. We test this on CFTC leveraged-fund and managed-money positions in 32 futures markets with a causal phase and a pre-registered design. Pooled, the order parameter predicts neither returns, drawdowns nor volatility of the speculative book, in or out of sample, and its excess over independence is 89% ordinary comovement of positions. Exploratory splits tell a different story: in currencies, synchronization precedes deeper drawdowns of the leveraged-fund book beyond aggregate positioning (-0.24 s.d., out-of-sample R^2 of 6.8%), while in commodities it precedes continuation. We register held-out tests of both.

Keywords: crowding, synchronization, Kuramoto model, Commitments of Traders, currency crash risk, pre-registration
JEL: G11, G13, G14, G17

Note on this version. This paper replaces the version posted on SSRN in March 2026 under the title "Endogenous Kuramoto Synchronization in Algorithmic Trading Flows". The empirical results of that version are withdrawn; Appendix B lists them with the reason for each.

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1. Introduction

Crowded positions unwind together (Pedersen, 2009; Stein, 2009; Khandani and Lo, 2011). Measures of crowding infer it from holdings, from return comovement (Lou and Polk, 2022), or from futures positioning, which also carries hedging pressure (Bessembinder, 1992; de Roon et al., 2000; Kang et al., 2020). In currencies, speculative net positions predict crash risk (Brunnermeier et al., 2008). Synchronization models from physics add a different object: map each trader group to a phase oscillator and measure alignment with the Kuramoto order parameter $r(t) = |N^{-1} \sum_i e^{i\theta_i(t)}|$ (Strogatz, 2000; Acebrón et al., 2005). High r is read as crowding.

This letter tests that reading on real positions. An earlier, unpublished version of this framework built phases from synthetic strategies with an event-anchored phase that interpolated to the next flow peak, hence used future data, and estimated a liquidity elasticity from a Corwin–Schultz spread, which is itself a function of the high–low range (Appendix A). We replace both with a causal phase on CFTC positions, pre-register the test, and report everything that followed: a registered null, a review that exposed why it is null, the exploratory sub-market effects behind it, and a registered held-out test of those effects.

2. Data and registered design

Positions are the weekly CFTC Commitments of Traders, futures only, Tuesday as-of: leveraged funds in the Traders in Financial Futures report (equity indices, Treasuries, currencies) and managed money in the disaggregated report (energy, metals, grains, livestock). The net position of market i is $x_i(t) = (\text{long} - \text{short})/\text{open interest}$. Archives were downloaded on 7 November 2026; histories before 2009–2010 are CFTC back-casts. A report enters from the first trading day on or after the Monday following its as-of date (it is released on Friday), and report weeks around the 2013, 2018–19 and 2025 shutdowns are excluded. Prices are back-adjusted daily futures; returns are daily price changes scaled by trailing volatility. The universe is every market with at least 85% coverage of both the CFTC series (2009–2026) and our price files (2010–2026): 32 markets. The price requirement removes the softs, palladium and KC wheat.

The phase is causal: with $z_i(t)$ the 156-week trailing z-score of x_i and $d_i(t)$ the weekly change of x_i divided by the trailing standard deviation of

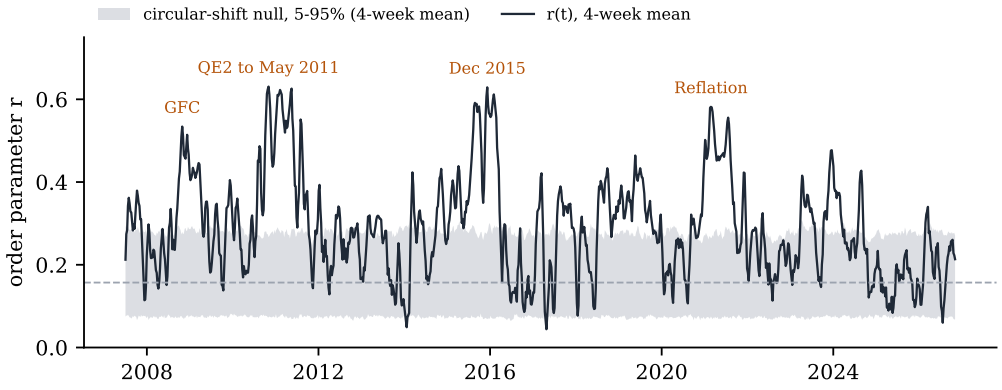


Figure 1: Order parameter of speculative positioning across 32 futures markets (causal phase, 4-week mean) against the pointwise 5–95% band of 500 circular-shift surrogates. Under the null about 5% of weeks exceed the band.

changes, $\theta_i(t) = \text{atan2}(-d_i, z_i)$, so that 0 is a maximal long and π a maximal short. Every phase at t is unchanged when data after t are altered (tested).

The registered hypotheses were: H1, mean r exceeds a null that circularly shifts each market's phase series ($p < 0.01$); H2, r predicts at least one of the 4-week return (Y1), maximum drawdown (Y2) or volatility ratio (Y3) of the speculative book (each market long or short with the sign of x_i, volatility-scaled, equal weight), with controls for the book's trailing return, trailing volatility ratio, pairwise correlation and first-component share of position changes, and mean $|z_i|$; stationary block bootstrap (Politis and Romano, 1994) and Romano–Wolf adjustment (Romano and Wolf, 2005); H3, out-of-sample improvement over the model without r (Clark and West, 2007), training to 2017 and testing on 2018–2026. A full crowding paper required all three. The registration (`PREREG.md`) and code were committed before each run; these timestamps are self-reported, not an external registration. A first run, affected by a loading bug, used only the 16 commodity markets; it is kept and reported in the package.

3. Registered results

H1 holds as registered: mean r is 0.287 against a surrogate mean of 0.156 ($p = 0.0005$, the smallest attainable with 2,000 surrogates), equal to the independence floor $\sqrt{\pi/4N} = 0.157$ (Figure 1). H2 and H3 fail (Table 1):

Table 1: Registered predictive tests, 32 markets (770 weeks, entries 2010-08 to 2026-01). b: coefficient on r in s.d. of the outcome, with Newey–West 95% interval; p_{RW}: Romano–Wolf adjusted; p_{CW}: Clark–West one-sided, 396 test weeks; MDE: minimum detectable effect, 80% power.

Outcome	b	95% interval	p_{RW}	p_{CW}	R^2_{oos}	MDE
Y1 return	0.04	[−0.07, 0.15]	0.78	0.40	0.000	0.16
Y2 drawdown	−0.09	[−0.20, 0.02]	0.33	0.21	0.005	0.16
Y3 vol. ratio	−0.01	[−0.11, 0.09]	0.78	0.82	−0.006	0.14

no outcome has an adjusted p below 0.33, and out-of-sample R^2 stays under 1%. The test could detect effects of about 0.15 s.d. of the outcome; for drawdowns the 95% interval reaches −0.20 s.d., so small crowding effects are not excluded.

4. Why the pooled test is null (exploratory)

The analyses in this section were run after the registered results and an adversarial review; they are hypothesis-generating.

H1 is mostly comovement.. The circular-shift null removes all dependence between markets, so any common factor rejects it. A multivariate Fourier surrogate of weekly position changes keeps every cross-correlation and spectrum and randomizes the rest (Prichard and Theiler, 1994). Against it, synchronization is marginal (Table 2): 89% of the excess of r over independence is linear comovement, as expected when traders respond to common shocks (Teramae and Tanaka, 2004). Only commodities keep a clear excess ($p = 0.006$). The order parameter is also close to a breadth measure: its rank correlation with $|N^{-1}\sum_i \operatorname{sign} z_i|$ is 0.80, and phases reduced to the sign of z_i give a mean r of 0.271 against 0.287.

The pooled null averages opposite effects.. Table 3 splits the registered H2 and H3. In currencies, higher r precedes deeper drawdowns of the leveraged-fund book (−0.26 s.d., $t = -4.85$), out of sample as well ($R^2_{oos} = 6.8\%$). In commodities, it precedes higher returns of the managed-money book (+0.18 s.d.), consistent with trend continuation (Moskowitz et al., 2012). Rates and equity indices show a drawdown effect too, but there leveraged funds are net short in most weeks in the longer Treasury contracts and the S&P 500

Table 2: Mean r against two nulls (500 surrogates each; complete-case panels, so the observed values differ slightly from the registered one). Share: part of the excess over the circular-shift mean that the Fourier null reproduces.

Markets	observed	circular mean	Fourier mean	p_{Fourier}	share
All 32	0.277	0.160	0.264	0.036	0.89
Commodities (16)	0.368	0.225	0.342	0.006	0.82
Financials (16)	0.306	0.233	0.298	0.194	0.89
Currencies (8)	0.424	0.315	0.417	0.253	0.93
Rates and equity (8)	0.427	0.329	0.404	0.052	0.76

Table 3: Exploratory split of the registered tests (registered controls). Last column: coefficient on r with aggregate positioning ($|N^{-1}\sum z_i|$, $N^{-1}\sum z_i$) added, t in brackets.

Markets	Outcome	b	t_{NW}	p_{CW}	R^2_{oos}	b with positioning
Currencies (8)	Y2 drawdown	−0.26	−4.85	< 0.001	0.068	−0.24 [−3.47]
Rates, equity (8)	Y2 drawdown	−0.28	−3.81	< 0.001	0.050	−0.13 [−1.48]
Commodities (16)	Y1 return	0.18	3.20	0.020	0.026	0.13 [2.14]

contract (UB 100%, ES 98%, ZB 84%, ZF 71%), positions that partly reflect basis trades and hedges of cash holdings rather than directional bets (Barth and Kahn, 2021); the effect there disappears once aggregate positioning is controlled. With $|N^{-1}\sum_i z_i|$ and $N^{-1}\sum_i z_i$ added, the channel of Brunnermeier et al. (2008), the currency effect stays (−0.24 s.d., $t = -3.47$, $p_{RW} = 0.001$) and so does commodity continuation (+0.13 s.d., $t = 2.14$, $p_{RW} = 0.034$). Five sub-universes and three outcomes make 15 tests. With the registered controls both effects survive a Bonferroni correction across them; with aggregate positioning added, only the currency effect does.

5. Registered held-out tests

All the effects above come from data already examined. We therefore register (`CONFIRM_PREREG.md`, to be deposited on OSF before any held-out outcome is computed) three one-sided tests on outcomes no analysis has looked at, each at $\alpha = 0.05/3$, with the registered controls plus aggregate positioning: CH1, currency r predicts deeper drawdowns of the currency book on entries from March 2026 to December 2028 (no price after 24 February 2026 has been loaded); CH2a, commodity r predicts higher returns of a managed-

money book on six commodity markets never used in any outcome analysis (cocoa, cotton, coffee, sugar, KC wheat, palladium); CH2b, the commodity effect on the time hold-out. The code refuses to run before the registration is recorded and the time hold-out before February 2029. Power is limited and stated in advance: 44% for CH1 at the in-sample effect and 13% at half of it, 17–30% for CH2a.

6. Conclusion

Speculative positions across futures markets move together, but most of that alignment is ordinary comovement, and pooled synchronization is not a crowding warning. The pooled null hides two opposite effects: synchronized leveraged-fund currency positioning precedes crash risk beyond the level of positioning, and synchronized commodity positioning precedes continuation. Both are exploratory and both are now registered for held-out tests.

Declaration of competing interest

The author declares no competing interests.

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Data availability

Code, both pre-registrations, every result file and the CFTC download script are at https://github.com/barj28/kuramoto-positioning-sync; an archived DOI will follow. CFTC data are public. Futures prices are licensed (Databento) and not redistributed; the weekly outcome and control rows are included.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) in order to write and test analysis code, run an adversarial review of the manuscript, and draft text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

CRediT authorship contribution statement

Arjun Kathiravelu: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing – original draft, Writing – review & editing.

Appendix A. Two flaws of the earlier framework

Event-anchored phases.. Assigning phase 0 to a buying burst, π to the next selling burst and interpolating in between needs the date of the next burst; peak detection that compares q_t with q_{t+1} adds a one-step leak. Altering a position series after date τ changes the interpolated phases before τ; the causal phase of Section 2 does not move.

A spread elasticity built from volatility.. The Corwin–Schultz spread is a function of the daily high–low range, which scales with volatility (Abdi and Ranaldo, 2017). Simulating a market whose true spread is constant (true elasticity zero) and regressing $\log(1/S_t)$ on log volatility gives elasticities of 0.71 to 0.87, close to the mean of 0.73 that the earlier framework reported across 43 CME futures; with volatility measured over the 21 days before t the estimates are 0.73 to 0.89.

Inference.. The earlier work reported i.i.d. p-values for correlations between a persistent $r(t)$ and overlapping 21-day forward drawdowns. Such p-values are known to be far too small (Hansen and Hodrick, 1980; Valkanov, 2003); all inference here is dependence-robust.

Appendix B. Claims of the March 2026 version that are withdrawn

1. The liquidity-elasticity spectrum across 1,543 instruments (1,500 US equities and 43 CME futures, from Corwin–Schultz spreads), its sector ordering, and the finding that Treasury-bill ETFs are the only sector with a negative elasticity. With a constant true spread, so a true elasticity of zero, the estimator returns 0.71 to 0.87 with contemporaneous volatility and 0.73 to 0.89 with the lagged 21-day volatility used for equities (Appendix A). The spectrum is not identified by this estimator.
2. The tick-level elasticities of 43 CME futures ($+1.49$ to -0.56) and the four-regime taxonomy built on them. Replacing contemporaneous realized volatility by its one-day lag, to remove bid-ask bounce, leaves a rank correlation of 0.24 with the reported estimates, and 24 of the 43 signs change.

3. The order parameter of 4,786 synthetic strategy–instrument positions and every result built on it: its correlation with forward drawdowns ($\rho = +0.196$), the March 2020 episode, the comparison with a rolling-correlation benchmark (paired $t = 3.04$) and the per-instrument cross-validations. The Signed Flux Bursts phase interpolated to the next flow peak and therefore used future data, the strategies were functions of prices rather than observed positions, and the drawdown correlations were tested with i.i.d. p-values on overlapping horizons.
4. The calibration of the mean-field model ($K_0 = 7.47$, $\gamma = -0.147$, $\delta = 8.45$): $\gamma < 0$ violates the model's own constraint $\gamma > 0$, and $\delta = 8.45$ contradicts the square-root impact exponent of 0.5 that the model assumes.
5. The endogenous coupling $K(r)$ and its consequences (stability regimes, the "liquidity black hole", hysteresis, capacity bounds) are not tested in this paper and are not maintained.

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